



Head-Up vs. Head-Down Displays: Examining Traditional Methods of Display Assessment While Driving

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ABSTRACT

Technological advances in Head-Up Displays (HUDs) have renewed vehicle manufacturer interest. Recent research links time with eyes off of the road to increased chance of accidents, a problem that could be diminished when using HUDs. Before HUDs can be safely integrated, there must be proven methods of assessing these displays. NHTSA established guidelines for assessing in-vehicle tasks; however, these have not been examined with HUDs, differ from traditional Head-Down Displays (HDDs). This study followed most of the NHTSA guidelines for the Eye Glance Test for a visual text search task while tracking driving and secondary task performance measures. The HUD performed worse on the NHTSA eye glance test than the HDD did; however, the driving performance measures were superior when driving with the HUD. There were no significant differences in the secondary task performance between the two displays. Therefore, the NHTSA standard may not adequately assess HUDs in vehicles.

Author Keywords

Display assessment; driving performance; head-down displays; head-up displays; secondary task performance

ACM Classification Keywords

H.5.2 Information Interfaces and Presentation (e.g. HCI): User Interfaces

INTRODUCTION

Driving is a task that requires a steady stream of visual input for the driver to adequately make decisions on vehicle positioning and evasive maneuvers. Even 2.0 seconds with a drivers' eyes away from the road can have devastating consequences as the drivers are not able to respond to the environments around them [1]. While there are many resources invested in educating the public about safe driving and minimizing distractions (e.g. distracted.gov),

drivers still engage in unsafe tasks, especially sending text messages from their cell phones [2]. Therefore, it appears that distracted driving caused by non-driving tasks will continue to be a concern in the future.

Distracting displays take drivers' visual attention away from the road entirely. While there are methods of ensuring that displays will not be overly distracting [3], poor timing of a distractor (e.g. cell phone, in-vehicle display) could still have disastrous effects if the driver missed an important visual cue while looking away from the roadway. With this in mind, there has been a renewed recent interest in Head-Up Display (HUD) incorporation into vehicles.

HUDs keep the visual stimuli on a display at or above the driver's line of sight in a vehicle. This minimizes the time spent switching between looking at a roadway and at a center console or dashboard display, which are considered to be Head-Down Displays (HDDs). HUDs may allow drivers to use peripheral vision to maintain vehicle stability even when not focused on the road ahead. Because HUDs are not yet widely used, limited research has been performed to assess the validity of common standards in regards to HUDs.

There are two primary reasons that HUDs might be beneficial to drivers: (1) visual attention remains in the direction of the road while in use, and (2) display integration with the world can provide more rich information and remove task uncertainty. The research in this paper focuses on the potential benefits of HUDs regarding visual attention. If visual attention when using a HUD remains such that drivers can use their periphery to assess the road while performing a task, then some potential problems could be alleviated.

Research on HUD use in vehicles is promising, with HUD use performing better in terms of vehicle control [4, 5], faster reaction to road events [4], faster response time for secondary tasks [6], and even lower levels of mental stress or workload [4, 6] as compared to HDDs. Other research exploring the impact of HUDs in driving scenarios has found minimal effects of HUDs on driving performance metrics [6, 7]. However, some researchers have found driving task decrements [8-10], as well as increased gaze durations away from the road [7]. Some of these performance decrements exist to a lesser degree as compared to HDDs [10], suggesting that HUDs may still be a viable option for future displays. The research on HUDs

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has returned mixed results with potential benefits but an incomplete picture of large-scale impacts and long-term implications of using these displays. Because of the mixed results regarding HUDs, they should be further studied, especially with respect to current driving display standards and guidelines [7, 8].

The National Highway Traffic Safety Administration (NHTSA) developed guidelines to assess driving displays [3]. The NHTSA visual-manual guidelines for vehicle displays provide a standard display assessment method that can be used in a variety of settings with a small sample size and modest equipment. These guidelines assess the probable distraction for a specific display by using either eye glance measurement in a driving simulator or occlusion testing [3]. This paper will compare a HUD to a more traditional HDD with respect to NHTSA eye glance measurement guidelines, driving performance, and secondary task performance. We will:

- I. Examine visual behavioral changes between HUDs and HDDs.
- II. Compare NHTSA task acceptance guidelines between the display types.
- III. Examine the impact of the display on other measures of both primary and secondary task Methods

Participants

Our participant pool met the guidelines specified by NHTSA [3]. Four gender-balanced age groups of 6 participants completed the study. The groups were separated into participants aged 18-24, 25-39, 40-54, and 55+. Participants needed to be healthy, actively drive, have experience using a cell phone while driving, and be unfamiliar with the devices used in testing [3]. However, due to population limitations, the requirement of driving a minimum of 3,000 miles per year was relaxed to driving regularly, with 5 participants driving less than 3000 miles per year and only 1 driving less than 1,000 miles (mean – 11,717).

Materials

Participants drove a fixed-base simulator with a National Advanced Driving Simulator (NADS) scenario displayed on a curved projection approximately 9 feet ahead with a 94° field of view. The cockpit of the vehicle was a left-hand drive 2014 Mini Cooper with all displays intact. For the HDD condition, a Surface Pro 3 tablet with a 12" screen was fixed to the right of the driver in front of the existing center console display, simulating a large-screen display. A Pioneer HUD with a fixed focal depth of about 3 meters was fixed above the driver's line of site for the heads up display (HUD) condition, per the manufacturer's recommendation (Figure 1).



Figure 1: Driving Simulator

The tasks were displayed to the participants using Microsoft PowerPoint with custom Visual Basic for Applications software. The software stored stimulus information and response times as the participant completed each trial. The same software was used for both displays and the Surface ran the software for both conditions (remotely for the HUD); however, the font size differed between the displays to ensure that the visual angle (VA) of the letters matched for both task conditions ($VA = 0.9$ degrees). Because of the slightly different viewing distances, the fonts differed in size to maintain this viewing angle (HUD = $0.375''$, HDD = $0.406''$). Both of our displays exceed visual-manual guidelines indicating that text should be larger than $0.25''$ for displays in vehicles (Figure 2) [11].



Figure 2: Display positioning from the driver's perspective

Participants were fitted with SensoMotoric Instruments (SMI) eye-tracking glasses, and iView ETG software was used to collect eye gaze data. Glances were analyzed using SMI BeGaze software. A Mio LINK electric heart rate (HR) monitor wristband was paired with an iPhone 6S running the Heart Rate Variability Logger application. Noldus software collected video data throughout each condition. Finally, the online survey service Qualtrics was used to collect NASA Task Load Index (NASA-TLX) data [12] and self-reported distraction and preferences. However, HR, video, NASA-TLX, distraction, and preference data were not analyzed for the current paper.

Primary Task

For each of the scenarios, participants followed a white lead vehicle that traveled at a constant speed of 65 mph. The scenario was a divided highway with no other vehicles except for the lead vehicle (Figure 3). Our roadway simulation was consistent with the majority of the recommendations made by NHTSA. For example, it was open and flat, made of asphalt, had 12 feet wide lanes with

shoulders, with solid white lines on the outside edges of the road [3]. The road had some curves but no data was collected during those times. Our four-lane roadway was divided and occasionally had additional lanes on either side of the driver, which differs slightly from the NHTSA recommendations. We are confident that these differences did not affect the primary driving task because the lead vehicle and participant stayed in the same lane. Since we are doing a car following task (e.g. as opposed to a lane changing task) these differences are not as apparent because the lead car did not change lanes.



Figure 3: For the majority of each drive, the driving scene was a straight 2-lane divided highway with only the white lead car.

While the NHTSA guidelines suggest driving in the right lane of an undivided 2-lane highway, these changes were necessary due to time constraints and limitations of our driving simulator. All data collected during the curve was excluded from the analysis. Thus, all data was collected only during times when the participant was following the lead vehicle on a straight section of road. Participants were instructed to follow the lead vehicle at a safe distance while maintaining a close enough distance to follow the vehicle with ease. No instructions regarding a specific distance or speed were given to participants.

Secondary Task

During two of the experimental driving scenarios, participants completed a secondary task composed of strings of random letters and numbers called “pseudo-text”. Figure 4 shows that there were multiple levels of complexity, and each line contained 30 randomly-ordered characters and generally followed ISO 9241-3 [13].

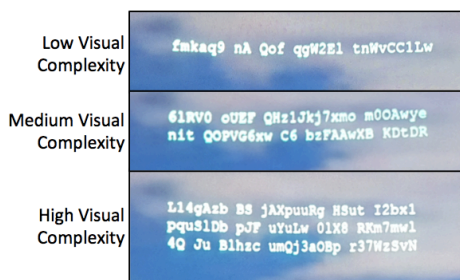


Figure 4: Drivers experienced three complexity levels, with 1, 2, or 3 lines of text on the presented image. Within one trial, both sets of text were the same level of complexity.

The secondary task needs to be sufficiently difficult to show differences between displays in regards to driving performance [6]. Therefore, the task included two distinct parts. First, participants searched through a string of text to find the character that was shown twice in a row. The double letter or number then became the target for the next trial (Figure 5A). This visual search section of the task did not require an exhaustive search of all text because participants were instructed that there was only one target character per image and because pattern recognition (e.g. looking for similar items grouped together) could be leveraged to increase speed. As soon as participants found the target, they could advance to the next portion of the task.

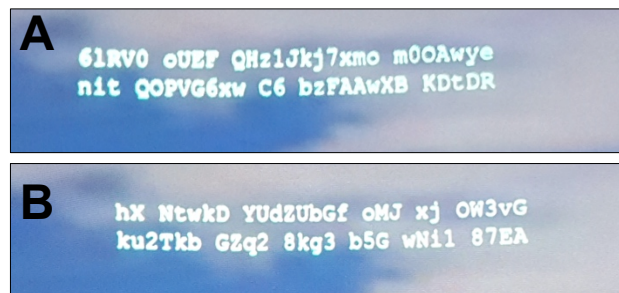


Figure 5: The target character from the first section would be “A” (on the second line). In the second segment, there is one instance of the target character “A”.

The second segment of the task required participants to scan through all of the text and search for the target character identified in the previous image and determine how many times it appeared (Figure 5B). The second image included at least one, but not more than seven targets; however, participants were not told anything about the potential number of targets in this image. To properly finish the task, participants had to check each individual character and determine if it matched the target. Checking each individual character requires participants to accurately track their place while remembering the count of target matches, and is not easily interruptible. The task was designed to tempt participants to look away from the road for as long as possible. This design was intended to provide a worst-case display scenario that would push drivers to their limits.

Minor exceptions were made to the ISO standard to ensure that (1) a target (double) character would definitely appear in the first image, (2) no additional double-characters would occur unintentionally in the first image, (3) the second image would have at least one occurrence of the target character, and (4) the second image would not include alternative versions (i.e. “E” when “e” is the target) of a target letter that might confuse the participant.

Procedure

Participants provided consent and completed a demographic questionnaire upon arrival. Participants were then introduced to the text search task on a tablet and allowed to practice it until they could complete tasks without asking

questions, consistently and correctly stated the target character, and felt comfortable that no further practice in the task would improve their performance.

Participants were then instructed on how to use the driving simulator and the researchers emphasized that above all else, participants should drive as safely as possible. They were given time to practice as much as they needed to feel comfortable driving the simulator with the researcher watching to ensure that their driving was stable. When they indicated that they felt comfortable driving, participants were asked to practice performing the text search task while driving as well. For each of these three actions (performing the task, driving, and driving while performing the task), participants were given an unlimited amount of time to practice until they felt that no further practice would improve their performance on any of the tasks. At that point, they were asked to begin the data collection.

Prior to beginning data collection, the researchers introduced the NASA TLX [12] and discomfort surveys to the participant and they completed a practice survey. Then, a wrist-worn heart rate monitor and the eye-tracking glasses were placed on the participant and calibrated.

Data was collected for 3 conditions for each participant, each of which required the participant to follow a lead vehicle: (1) a baseline drive with no secondary task, (2) driving while completing the secondary task on a HDD, and (3) driving while completing the secondary task on a HUD. These conditions were counterbalanced across each age group and across genders. Prior to each drive, the display to be used was turned on and participants were informed which display would be in use for that condition. Each drive began with the participant shifting the simulator car into drive and beginning to follow the lead car. After they were driving while maintaining sufficient speed and lane position, the researchers initiated the secondary tasks if applicable to the condition.

A bell sound indicated the beginning of an individual trial so participants would know to start searching for the target character on the previously specified display. After locating the target, participants verbally stated the target character and proceeded to the next text string by pressing any button on the steering wheel (Figure 6).



Figure 6: Participants could advance the task by pressing any of the steering wheel buttons (right and left, circled above).

The researchers recorded the named target for analysis. As soon as the button was pressed, a bell sound indicated that the new text was available on the screen and participants could begin to count how often their target appeared. Once they completed their search, participants pressed any button on the steering wheel to end the task, and verbally stated how many times the target occurred in the second image as well as their confidence on a scale of 0-10 (where 10 indicates that they were confident that they performed the task perfectly). These two numbers were also immediately recorded by the researchers. Between trials, there was an 8-12 second window of time where the participants did not have any task to attend to and could focus solely on driving. The goal of this break from the task was to minimize effects from a previous trial on a subsequent trial's driving performance. Each participant completed 5 replications of the 3 levels of complexity mentioned earlier for a total of 15 randomized trials in a single driving condition.

Upon completion of all 15 trials, the participant stopped the vehicle and the simulation was turned off so that the participant could complete the NASA TLX and discomfort surveys. This process was repeated for each driving condition and after all drives, participants completed additional surveys regarding NASA-TLX, display preferences, and perceived distraction. After completing all surveys, participants were compensated and thanked for their time.

ANALYSIS

A significant amount of data reduction was required to eliminate extraneous data. All 24 participants completed all 15 trials under each display condition ($24 \times 15 \times 2 = 720$ data points). All driving and eye tracking data and the corresponding secondary task measures occurring during a curve in the driving scenario were immediately excluded from any further analysis (51 trials lost). Next, driving and eye tracking data were parsed into segments corresponding to individual tasks and each trial was analyzed individually. In some cases, an enthusiastic participant pressed the button with such zeal that it skipped the following text string. The data for these trials was incomplete and therefore was also excluded from analysis (24 trials lost). The remaining trial data ($N=645$) was used for HDD and HUD analyses.

Because there were no trials during the baseline period, five 20-second nonconsecutive driving segments were analyzed beginning after the first minute of the baseline drive. This allowed enough time for the driver to settle into the drive without overlapping with the curve. These segments were compared to the driving segments while participants were completing a trial. Reducing the data in this method allowed us to analyze all measures at the display-complexity level (e.g. HUD-High) to explore the impacts of complexity in addition to display type.

Driving Measures

Single-point driving data was reported every $1/60^{\text{th}}$ of a second. The driving data for each trial was analyzed

individually for longitudinal and lateral control according to SAE J2944 10.1.1.3 (option C). The lateral control of the vehicle was measured by *Standard Deviation of Lane Position* (SDLP) where the 0 point corresponded to the center of the lane with the lead car in it. The longitudinal control of the vehicle was measured by *Standard Deviation of Time Headway* (SDTH) and *Standard Deviation of Speed* (SDS). The time headway between the lead car and the participant's vehicle was tracked over time. The standard deviations of the reported lane positions, time headway values, and speed were calculated to determine SDLP and SDTH.

Eye Tracking Measures

The BeGaze software used for analysis tracks fixations, saccades, and blinks with respect to time and location. To convert these into glances, the duration of a saccade leading to an area of interest (i.e. HUD, HDD, or roadway) was summed with the consecutive saccades and fixations within that same area of interest. Blinks in an area of interest were considered to begin a new glance. The duration of all gazes away from the roadway for a single trial were summed to a single value reported as the *total eyes off road time* (TEORT). Any glance away from the roadway longer than 2.0 seconds was counted and reported for each individual trial in the *long glance count* metric. Finally, the *average glance duration* of each glance at a specific area of interest was averaged for individual trials.

NHTSA Guidelines

The NHTSA eye glance measurement test determines task acceptance based on three acceptance criteria requiring at least 21 out of 24 participants to succeed for a task to "pass": (1) no more than 15% of the eye glances away from the road should last longer than 2.0 seconds, (2) the average duration of all glances away from the road should be less than 2.0 seconds, and (3) the total time that participants spend with their eyes off of the road cannot exceed 12.0 seconds for one trial [3]. These calculations were all made at the individual level, with averaged values for the display-complexity level.

Secondary Task Performance

Task performance was evaluated with respect to both response time and errors. The response time, or the *total time* to complete a task was evaluated at the trial level by tracking the beginning and end time for each trial. There were two types of error metrics examined. First, the identified target character was checked for correctness and the *target accuracy* was recorded as a binary success or failure. Next, the number of targets counted was subtracted from the actual number of targets that occurred in the second string of text. This *target count* provided a signed number indicating over- or under-estimation of targets present.

RESULTS

Due to the small sample size and paired data from the within-subjects design, nonparametric methods (Wilcoxon Signed Rank) were used for all data analysis, at 0.05 level of significance.

Driving Performance Data

Lateral Driving Performance

When using the HDD, the SDLP during both the medium and high complexity levels were significantly different from both the HUD condition and the baseline. With respect to SDLP, we found a significant difference between the HDD condition with the high complexity visual task (HDD-High) and the HUD condition with the high complexity visual task (HUD-High) ($p=.010$) and between HDD-High and the baseline ($p<.001$). We also found a significant difference between HDD-Med and HUD-Med ($p=.008$) and between HDD-Med and the baseline ($p=.020$). The SDLP was consistently lower when using the HUD as compared to the HDD (Figure 7). The SDLP of the HUD conditions were not significantly different from the baseline (High: $p=.253$; Med: $p=0.877$; Low: $p=.427$). The SDLP for the low complexity in the HDD condition was also not significantly different from the HUD condition or the baseline ($p=.543$ and $p=.279$, respectively).

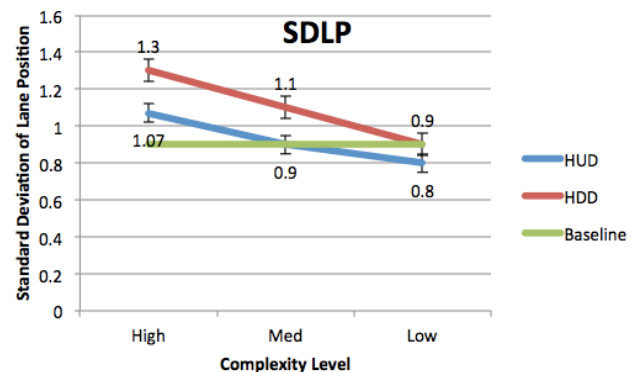


Figure 7: SDLP for each condition

Longitudinal Driving Performance

At the more difficult complexity levels, both displays show some driving performance decrement with regards to SDTH when compared to the baseline (Figure 8). When compared to the baseline, the HUD-High, HDD-High, and HDD-Med all exhibited significant differences ($p=.002$, $p<.001$, $p=.020$ respectively). The HUD-Med, HUD-Low, and HDD-Low did not exhibit any significant differences from the baseline ($p=.348$, $p=.464$, $p=.585$ respectively). The SDTH for the HUD conditions was not significantly different from the HDD conditions (High: $p=.236$; Med: $p=.146$; Low: $p=.910$).

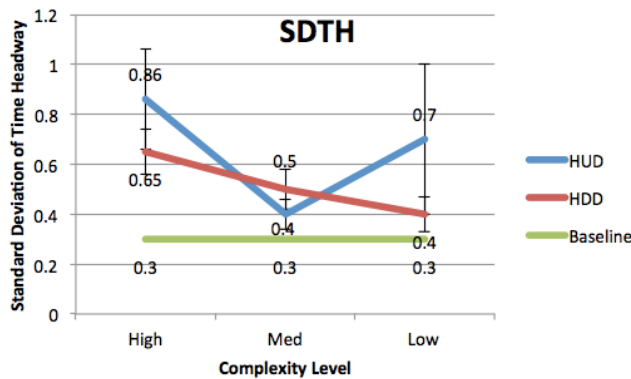


Figure 8: SDTH for all conditions

The SDS was only significantly different between the low HDD condition and the baseline drive ($p=.007$). All other comparisons to the baseline or across display types were not significant (Figure 9).

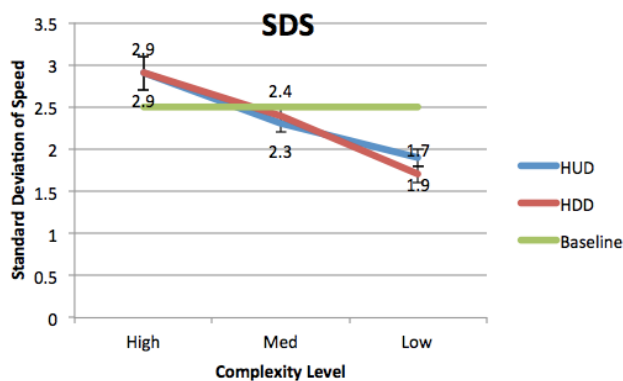


Figure 9: Standard Deviation of Speed

Eye Tracking Measures

Total Eyes Off-Road Time (TEORT)

The total time participants spent with their eyes off-road differed only in the medium level of complexity (Table 1).

Comparison for Wilcoxon		p-value	Z
HDD-High	HUD-High	.516	-.649
HDD-Med	HUD-Med	.030	-2.175
HDD-Low	HUD-Low	.203	-1.273

Table 1: TTEOR Wilcoxon Rank Sum Results

Long Glance Count

The number of long glances (>2.0 sec) differed between the HUD and HDD across all levels of complexity ($p<.001$ for each pair). Figure 10 shows the mean glance count for a given complexity level and display.

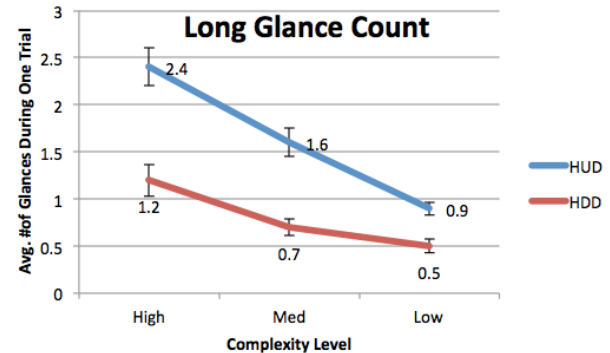


Figure 10: There may be an interaction effect between complexity level and long glance count.

Average Glance Duration

The average duration of a glance at a display differed between the HUD and HDD across all levels of complexity ($p<.001$ for all). There was no significant difference between any of the HDD complexity levels when compared to other complexity levels on the same display (High-Med $p=.593$, High-Low $p=.731$, Med-Low $p=.461$). For the HUD, there was not a significant difference between the High and Medium complexity levels ($p=.146$), but there was a difference between High and Low ($p=.001$) and between Medium and Low ($p=.043$). Figure 11 shows the mean glance durations for each display-complexity level.

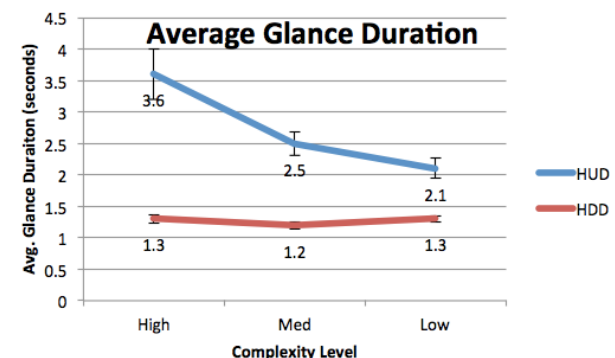


Figure 11: There is a strong possibility of interaction effect between complexity level and display.

NHTSA Guideline Metrics

Condition	<15% Long Glances (>2.0 sec)	Avg. Glance Duration <2.0 sec	TTEOR <12.0 sec
HDD-High	15	21	0
HUD-High	6	9	1
HDD-Med	15	21	11
HUD-Med	6	10	14
HDD-Low	16	22	22
HUD-Low	5	13	22

Table 2: Number of participants who meet the NHTSA guidelines requirements. 21/24 participants must succeed for a task to pass.

Secondary Performance Measures

Total Task Time

The total time spent completing the task showed significant differences at the high ($p=.016$) and medium ($p=.014$), but not the low ($p=.210$) levels of complexity (Figure 12).

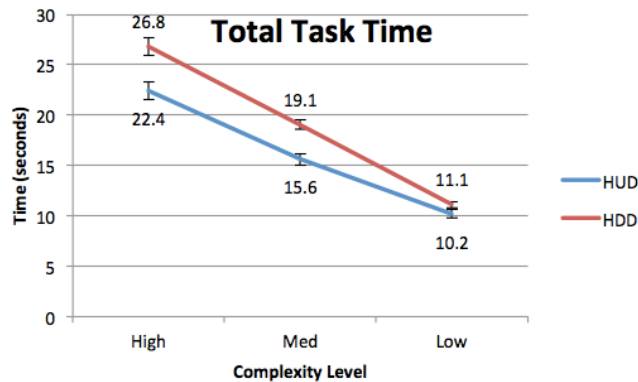


Figure 12: Total Task Time in Seconds to Complete a Trial

Error Measures

Target accuracy for each display was compared and there were no significant differences (High: $p=.281$; Med: $p=.599$; Low: $p=.539$). Similarly, the target count error was compared across all levels of complexity, but there were no significant differences (High: $p=.695$; Med: $p=.509$; Low: $p=.640$).

DISCUSSION

Driving metrics

The tasks were sufficiently difficult to result in a significant impact on driving, as seen by the effect of the tasks on both lateral and longitudinal variation. However, there was no significant difference between the HUD and the baseline drive lateral variability at any level of complexity. The lack of significant difference between SDLP of HUD and baseline suggests that drivers are able to adequately maintain lateral position while performing these tasks using the HUD. Thus, lateral driving was more negatively impacted when performing tasks in the HDD condition.

Some decrement in longitudinal vehicle control was observed in the HUD-high and HDD-medium and high conditions. Again, it is clear that this task was ultimately able to force a task decrement. For both lateral and longitudinal measures, the HDD was more often detrimental to drivers. This indicates that task complexity was difficult enough to cause change and shows greater impacts when driving with a HUD. The results from this study reflect prior research showing that HUDs result in less severe driving performance decrements than HDDs [10].

Eye Glance Measures

There were significantly different eye glance tendencies across displays. The HUD seems to have increasing impact on both long glance count and average glance duration as complexity increases. This suggests a different visual

response to HUDs. The drivers maintained similar glance duration with the HDD, but the HUD resulted in significant differences. Therefore, the HUDs more negatively affected glance patterns.

NHTSA Guideline Metrics

A main purpose of this paper was to examine how HUDs performed when tested with the current NHTSA guidelines for acceptance. However, the task was designed to tempt users to look away from the road for as long as possible. This was to make the task as difficult as possible and exaggerate differences between the two display types. One side effect of this design is that because the second portion of the task was not at all resumable, the participants “failed” the NHTSA guidelines.

However, the HDD passed the test for average glance duration for all three complexity levels. For the long glance percentage, the HDD showed a significant improvement over the HUD even though it still did not meet the requirements. We know the task is not safe for driving, and designed it as such, but what is interesting here is that the HDD consistently scores better with regards to the NHTSA guidelines.

Secondary Task Performance

There were no differences in task accuracy when using a HUD as compared to a HDD. However, participants were able to complete the tasks more quickly when using a HUD. Therefore, they were more efficient while maintaining accuracy. In driving, this would be of interest because quickly finishing distracting tasks allows drivers to get back to the primary task of driving more quickly.

CONCLUSION & FUTURE WORK

Overall, participants using a HUD maintained driving stability while performing the tasks more quickly and equally accurately as compared to a HDD. However, the HUD regularly scored worse on the NHTSA guideline metrics and HUD use more drastically impacted glance patterns than the HDD. These long glance durations would typically be associated with degraded driving performance, and yet the degradation did not correspond to the NHTSA performance. The significant difference in eye glance patterns without a corresponding decrement in driving and task performance suggest that people use visual search methods on HUDs in a fundamentally different way than traditional HDDs. These findings suggest that the NHTSA standards using eye glance patterns as a method of validating tasks may be skewed to reject HUDs even when HUDs are potentially safer than HDDs for a given task. This suggests that current guidelines should be re-evaluated with HUDs in mind. Perhaps HUDs should be governed by a different set of guidelines that are not so closely tied to glance patterns.

While long glances to a HUD may not have a large impact on positional vehicle control, this does not mean that the displays are inherently safer. It is possible that the displays

simply need to be assessed in a different way to understand the impact of HUDs on drivers. Because glance duration does not appear to be the key factor for successfully maintaining lateral and longitudinal lane position, perhaps a future standard for in-vehicle HUDs should incorporate situational awareness measures to better understand implications of HUDs for driving. The driving task used in this study had no surprise events or hazards in the roadway. While the HUD seemed to have superior driving and task performance, future research should examine the role of situational awareness in driving with HUDs.

One limitation of this study is the fact that we looked only at one type of task, and this particular task may not fully examine the possible differences between displays while driving. Similar studies that looked at spatial tasks or mixed-mode tasks might reveal more significant differences between the driving and NHTSA performance. More studies should be done to extend this work with a variety of task types.

An extension of this work would be to examine the data for each portion of the task independently. The first part was more interruptible and supported scanning while the second required individual character assessment. There appears to be an interaction regarding complexity. Perhaps the type of task has different implications with different displays as well. This could give us insight into whether different types of tasks have different resulting glance patterns for HUDs as compared to HDDs. Regardless, more work must be done before HUDs are widely implemented in vehicles.

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